

GIESSEN, Germany

WMO: 105320

Lat: 50.6017N

Lon: 8.6439E

Elev: 204

StdP: 98.90

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-12.0	-9.2	-15.0	1.0	-10.2	-12.5	1.3	-8.0	10.6	6.3	9.1	5.8	2.3	20	0.527

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	29.6	19.2	27.8	18.2	26.1	17.4	20.2	26.7	19.2	25.6	18.4	24.2	3.2	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.0	13.2	22.1	17.0	12.4	21.5	16.1	11.7	20.7	58.8	26.9	55.5	25.8	52.6	24.3	23.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.0	6.9	6.1	-12.6	33.0	4.4	1.3	-15.7	33.9	-18.3	34.7	-20.8	35.4	-24.0	36.3		
(5)			-12.9	22.3	4.4	0.7	-16.0	22.8	-18.6	23.2	-21.1	23.6	-24.3	24.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.5	0.8	0.7	5.1	8.4	13.3	16.3	18.7	18.4	14.5	10.0	4.6	2.4		
	DBStd	7.42	5.20	4.56	3.49	3.48	3.60	3.50	3.16	3.01	2.93	3.38	3.63	4.05		
	HDD10.0	1227	286	261	153	71	11	1	0	0	1	42	164	237		
	HDD18.3	3342	544	494	409	298	159	81	36	38	119	258	413	495		
	CDD10.0	1038	0	1	3	23	113	189	268	259	137	42	2	1		
	CDD18.3	111	0	0	0	0	3	19	46	38	4	0	0	0		
	CDH23.3	1153	0	0	0	1	61	193	469	379	49	1	0	0		
	CDH26.7	285	0	0	0	0	4	36	133	107	6	0	0	0		
(14) Wind	WSAvg	2.6	3.1	2.8	2.9	2.8	2.5	2.5	2.4	2.3	2.2	2.4	2.6	2.6		
(15) Precipitation	PrecAvg	734	64	52	49	45	62	67	76	64	59	62	65	70		
	PrecMax	930	135	133	105	93	118	135	151	145	134	182	107	191		
	PrecMin	574	6	6	9	2	14	28	32	6	23	10	5	17		
	PrecStd	90	31	30	26	21	28	28	29	31	25	35	24	32		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.6	13.9	18.5	22.5	27.1	29.8	31.8	32.5	27.5	21.7	14.0	13.5		
		MCWB	9.3	10.2	11.1	13.0	16.3	19.7	19.4	19.4	17.2	15.1	11.4	10.7		
	2%	DB	9.6	10.8	14.7	19.6	25.0	27.6	29.9	29.5	24.7	18.8	12.1	10.8		
		MCWB	7.6	8.4	9.3	11.7	15.1	18.2	19.4	18.8	17.0	14.2	10.3	9.1		
	5%	DB	8.4	8.4	12.2	17.5	23.3	25.6	28.1	27.3	22.5	16.6	10.8	9.4		
		MCWB	6.8	6.3	8.2	10.7	14.5	17.1	18.6	18.0	16.2	13.3	9.1	8.2		
	10%	DB	7.1	6.5	10.5	15.1	21.1	23.6	26.1	25.2	20.2	15.0	9.5	8.0		
		MCWB	5.8	4.7	7.6	9.3	13.8	16.5	17.8	17.1	15.0	12.1	8.0	6.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.5	10.5	11.9	13.7	17.6	21.1	21.6	21.1	19.2	16.4	11.8	10.7		
		MCDB	11.1	13.3	15.7	20.9	23.7	26.9	28.9	29.0	24.2	20.0	12.5	12.5		
	2%	WB	8.1	8.5	10.1	12.5	16.3	19.6	20.3	19.7	17.8	14.9	10.7	9.5		
		MCDB	9.4	10.3	13.7	17.9	22.7	25.3	27.3	26.2	22.3	17.6	11.9	10.7		
	5%	WB	7.0	6.5	9.0	11.2	15.3	18.2	19.4	18.8	17.0	13.7	9.3	8.1		
		MCDB	8.1	8.1	11.8	16.2	21.5	23.6	26.3	25.1	21.1	16.0	10.5	9.2		
	10%	WB	5.8	4.9	7.8	10.1	14.3	17.1	18.4	18.0	15.9	12.6	8.2	7.0		
		MCDB	7.1	6.3	10.3	14.6	19.9	22.1	24.6	23.7	19.3	14.6	9.3	7.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.5	6.3	7.5	9.3	10.4	9.6	10.7	10.7	9.3	7.5	5.1	4.2		
		MCDBR	5.2	8.3	11.2	13.5	14.6	13.6	14.3	14.7	13.3	9.8	5.9	5.7		
	5% WB	MCWBR	4.6	6.1	6.8	6.9	6.8	5.9	6.1	5.8	6.6	5.8	4.7	4.9		
		MCWBR	5.2	7.9	9.8	12.0	12.7	12.2	12.5	12.1	11.1	8.7	5.5	5.5		
(40) Clear-Sky Solar Irradiance	taub		0.315	0.340	0.375	0.402	0.406	0.407	0.415	0.410	0.379	0.364	0.339	0.312		
		taud	2.400	2.341	2.276	2.251	2.268	2.283	2.258	2.300	2.370	2.394	2.397	2.393		
	Ebn at Noon		715	785	819	838	851	851	836	821	808	747	673	656		
		Edh at Noon	63	85	108	124	128	127	128	117	98	80	63	55		
(44) All-Sky Solar Radiation	RadAvg		0.75	1.45	2.57	4.10	4.84	5.36	5.08	4.37	3.14	1.83	0.83	0.56		
	RadStd		0.11	0.22	0.36	0.51	0.53	0.51	0.62	0.40	0.43	0.20	0.11	0.06		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (47 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page